

ACF and PACF, Learned by Example

How a time series remembers — AR(1) and MA(1)



Five steps to reading the two plots

01	————	Basics	Series · correlation · lag Monthly prices · scatter · shift a row
02	————	ACF and PACF	Total vs direct resemblance Grandpa analogy · formulas · reading plots
03	————	AR(1)	Last month's price carries on Dominoes · ACF decays · PACF cuts off
04	————	MA(1)	Last month's shock lingers Shared shocks · ACF cuts off · PACF decays
05	————	Picking a model	Decide from the plots Comparison · quiz · Python

Goal: tell AR from MA by looking at just two plots

01

Three basic ideas

Time series · correlation · lag

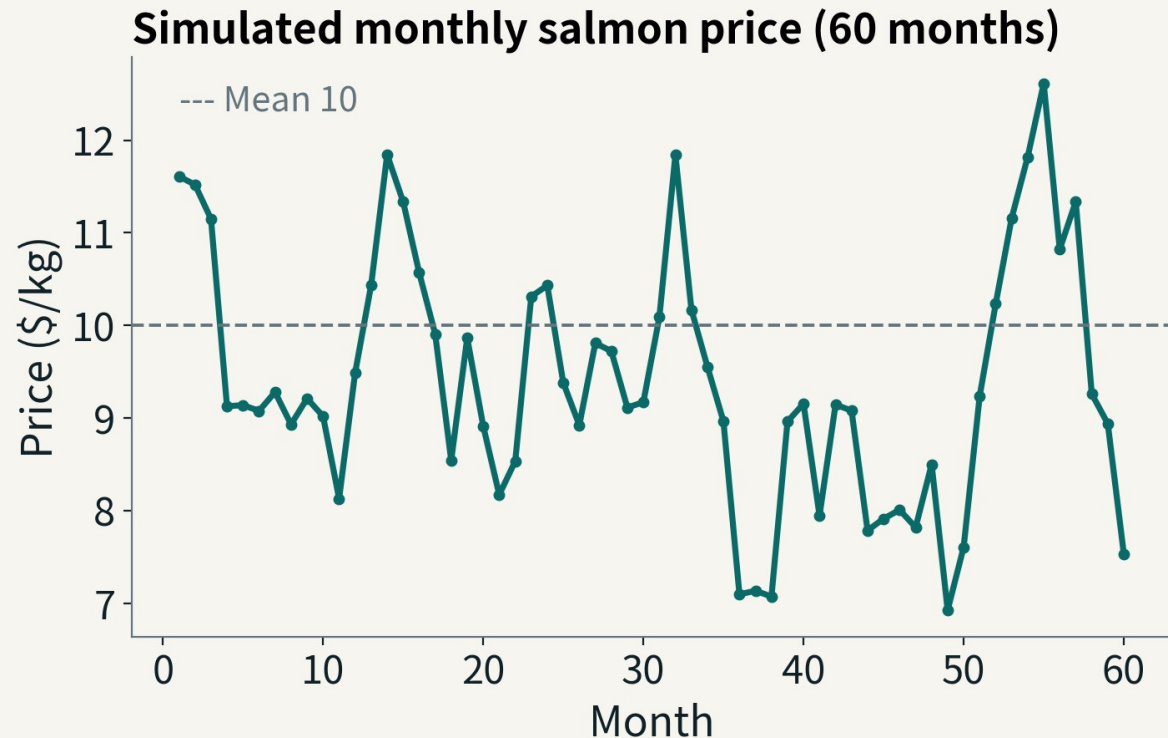
A time series is numbers recorded in time order

EXAMPLE

Salmon prices

- One value a month
- Order matters
- Close to last month

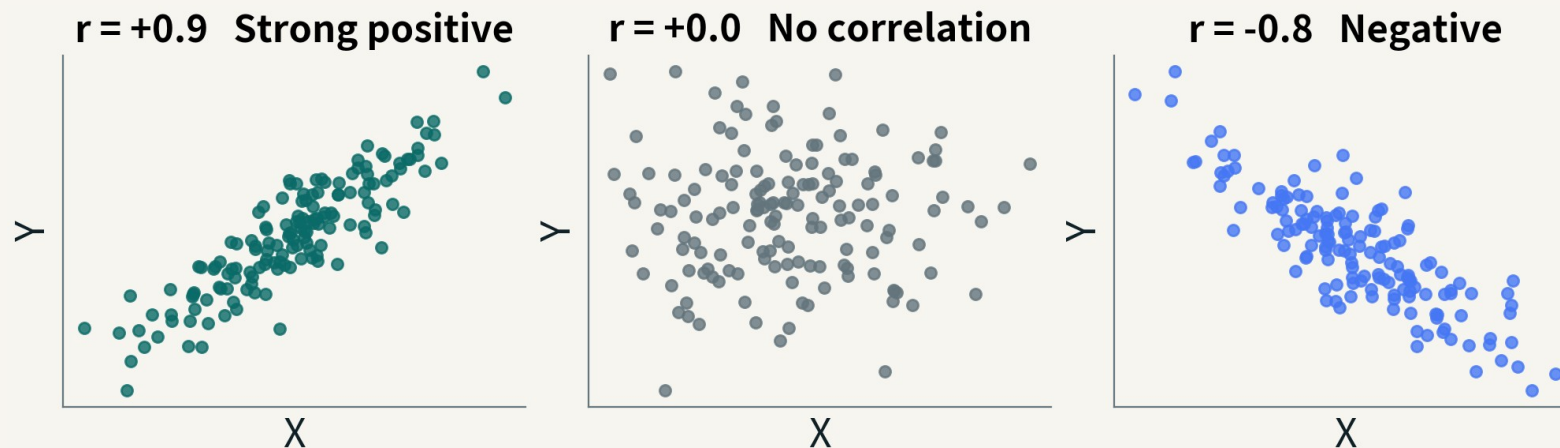
Today resembles
yesterday



Key question: how much does the past explain this month?

Correlation r measures how two numbers move together

$$r = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$



r is between -1 and $+1$ · $+1$ moves together, 0 unrelated, -1 opposite

A lag is made by shifting the table one row

Month	S_t This month	S_{t-1} 1 month ago	S_{t-2} 2 months ago
Mar	11.1	10.8	10.2
Apr	10.5	11.1	10.8
May	9.9	10.5	11.1
Jun	10.4	9.9	10.5

Shift the column down one row to make a lag column

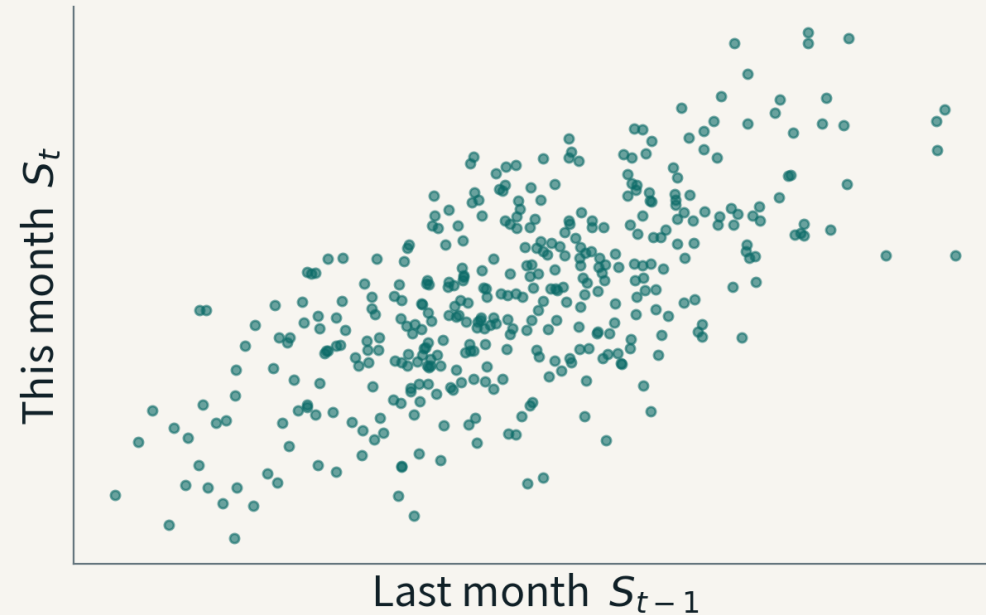
Lag 1: correlate these two columns $\rightarrow \rho_1$ Lag 2: correlate these two columns $\rightarrow \rho_2$

Correlation between a series and its shifted copy = autocorrelation

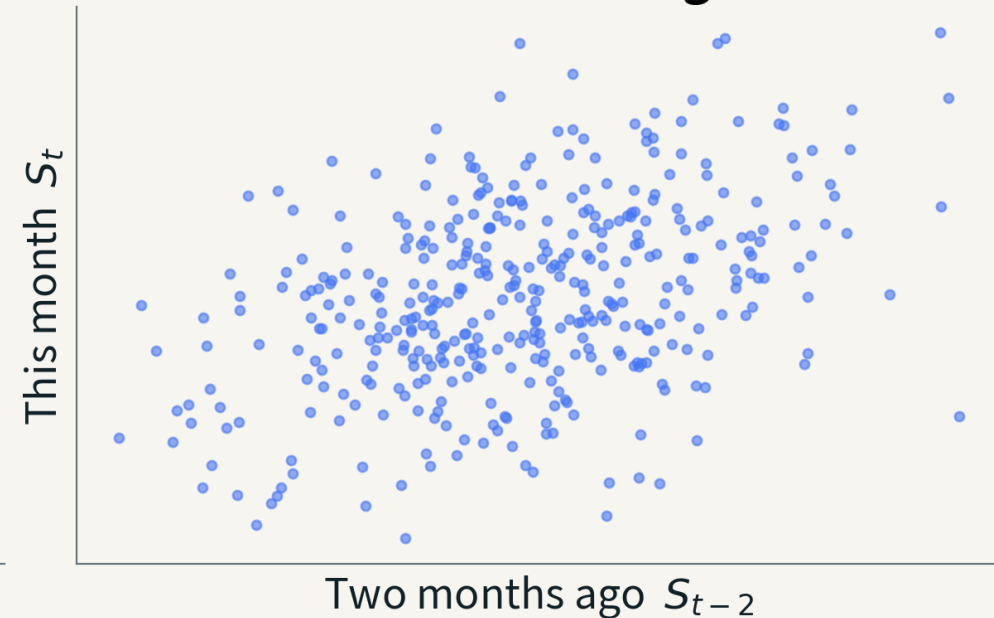
Example: lag-1 correlation beats lag-2

Scatter of 400 simulated monthly prices — one dot per month

This month vs last month: $r \approx 0.66$



This month vs 2 months ago: $r \approx 0.40$



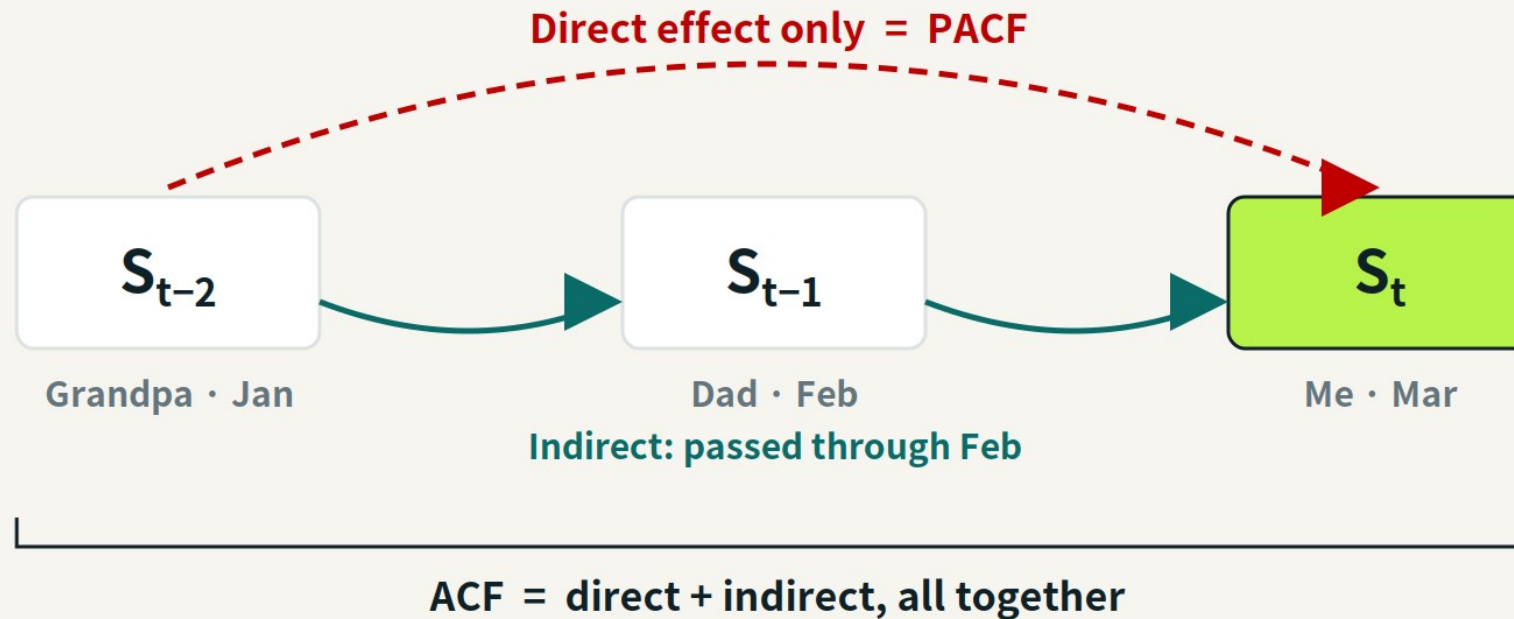
Collect r for each lag and plot them as bars: that is the ACF

02

ACF and PACF

Total resemblance vs direct resemblance

Two months ago reaches this month by two paths



ACF counts both paths

PACF counts only the direct path

Think of heights: Grandpa, Dad, and me

ACF

Plain resemblance

- Compare me and Grandpa
- Includes Dad's route

Indirect effects included

hold Dad fixed



PACF

Direct resemblance

- Only families, same Dad
- Resemblance left over

Pure effect, indirect removed

PACF = correlation that remains after removing the in-between months

ACF is the correlation between prices k months apart

Definition

$$\rho_k = \text{Corr}(S_t, S_{t-k})$$

Example: lag 2 = pairs two months apart

$$\rho_2 = \text{Corr}(S_{\text{Mar}}, S_{\text{Jan}}), \text{Corr}(S_{\text{Apr}}, S_{\text{Feb}}), \dots$$

Step k

k = 1: one month apart

k = 2: two months apart

k = 3: three months apart

Repeat for k = 1, 2, 3 ... and draw one bar per lag

PACF is the direct correlation left after removing middle lags

Regression view: the boxed coefficient

$$S_t = \phi_{21} S_{t-1} + \boxed{\phi_{22}} S_{t-2} + e_t$$

HOW TO READ

With last month in the equation, two months ago keeps only the extra explanation it adds.

Computed from correlations

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

THE NUMERATOR

Squared lag-1 = the two-hand-off guess. Its gap to the real lag 2 is the direct effect.

At lag 1 there is nothing in between, so PACF = ACF

Let's compute a PACF with real numbers

Example A · lag-1 0.8, lag-2 0.64

$$\phi_{22} = \frac{0.64 - 0.8^2}{1 - 0.8^2} = \frac{0}{0.36} = 0$$

Example B · lag-1 0.5, lag-2 0.50

$$\phi_{22} = \frac{0.50 - 0.5^2}{1 - 0.5^2} = \frac{0.25}{0.75} \approx 0.33$$

READING A

No direct effect

0.64 equals the hand-off 0.8×0.8

READING B

A direct effect exists

0.50 is larger than the hand-off 0.25

Real above the hand-off → positive PACF; below → negative

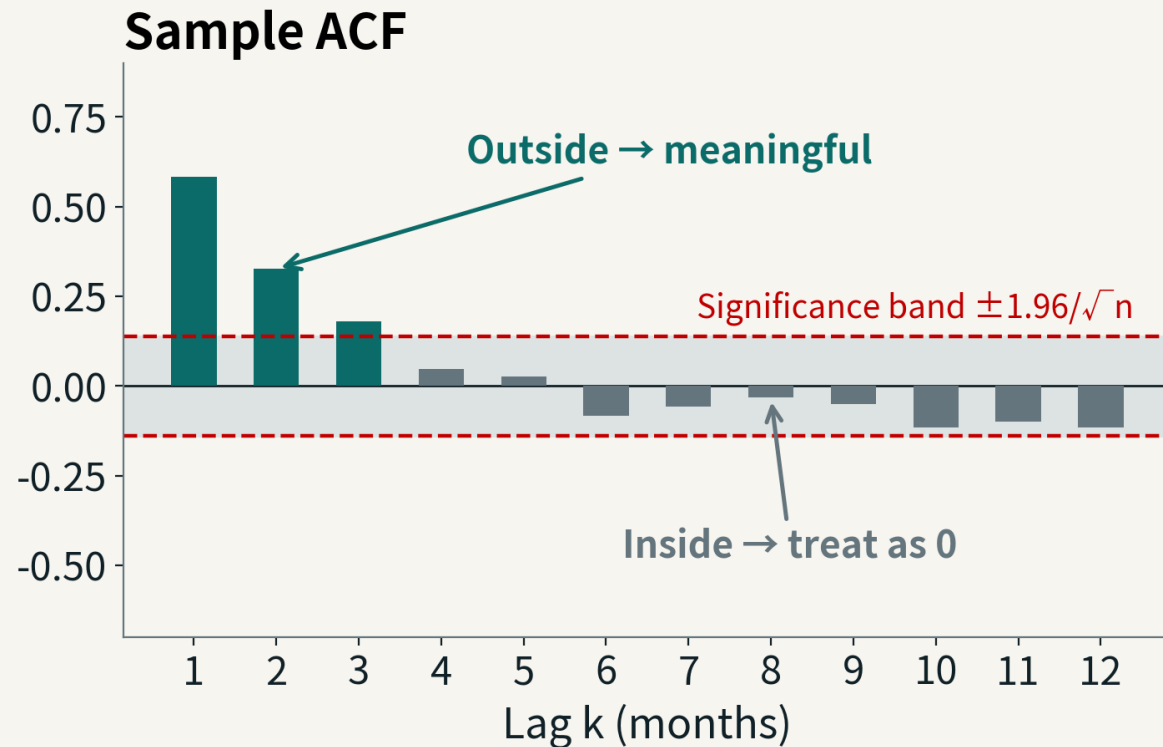
In the plot, only bars outside the band matter

READING THE PLOT

Bars and band

- Bar = correlation
- Band = $\pm 1.96/\sqrt{n}$
- Outside = real

$n = 200 \rightarrow \text{about } \pm 0.14$



Small bars inside the band are treated as zero (just noise)

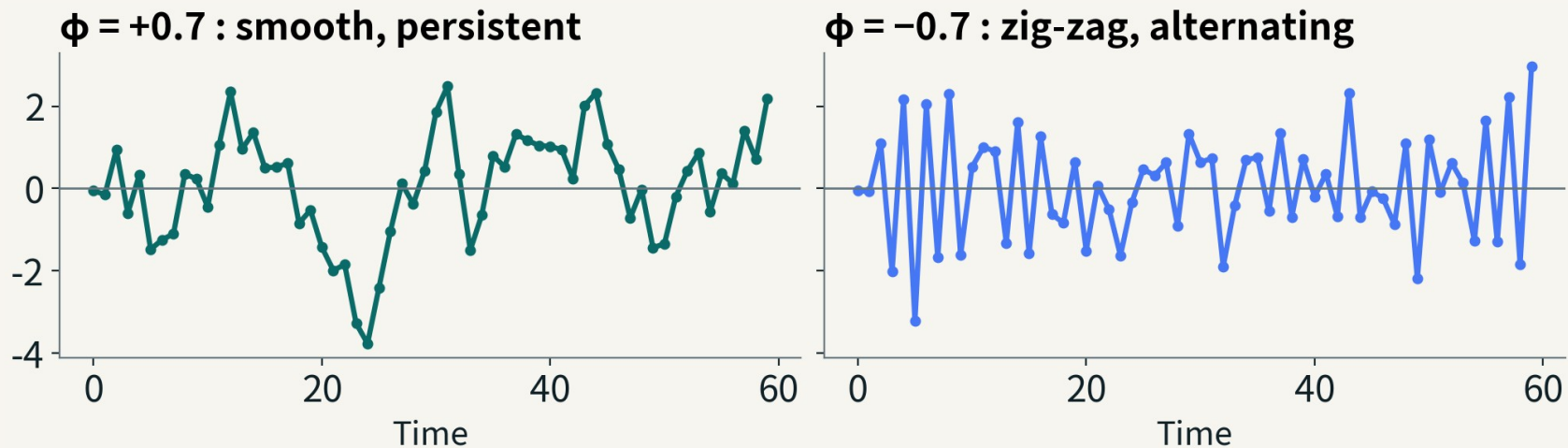
03

AR(1): last month's price carries on

AutoRegressive — explained by its own past values

In AR(1), part of last month's price carries into this month

$$S_t = \phi S_{t-1} + \varepsilon_t, \quad |\phi| < 1$$



Example: $\phi = 0.7$ keeps 70% of last month's deviation

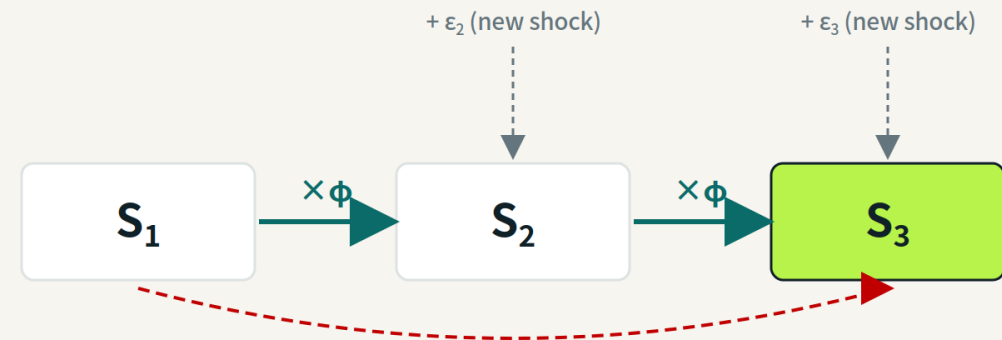
AR(1)'s ACF shrinks slowly, one ϕ at a time

Formula

$$\rho_1 = \phi, \quad \rho_2 = \phi^2, \quad \rho_k = \phi^k$$

Example

$$\phi = 0.7 : \quad 0.70, 0.49, 0.34, 0.24, \dots$$



Jan $\rightarrow$ Mar effect = $\phi \times \phi = \phi^2$

Feb's price already carries Jan's influence
Once Feb is known, Jan adds nothing $\rightarrow \text{PACF}(2) = 0$

If $\phi = -0.7$, signs alternate: $-0.70, +0.49, -0.34 \dots$

Effects pass on like dominoes, so the ACF never hits zero

AR(1)'s PACF is zero from lag 2 on

Plug in: the numerator is zero

$$\phi_{22} = \frac{\phi^2 - \phi^2}{1 - \phi^2} = 0$$

WHY

Feb already holds Jan's effect

With Feb fixed, Jan adds nothing new

RESULT

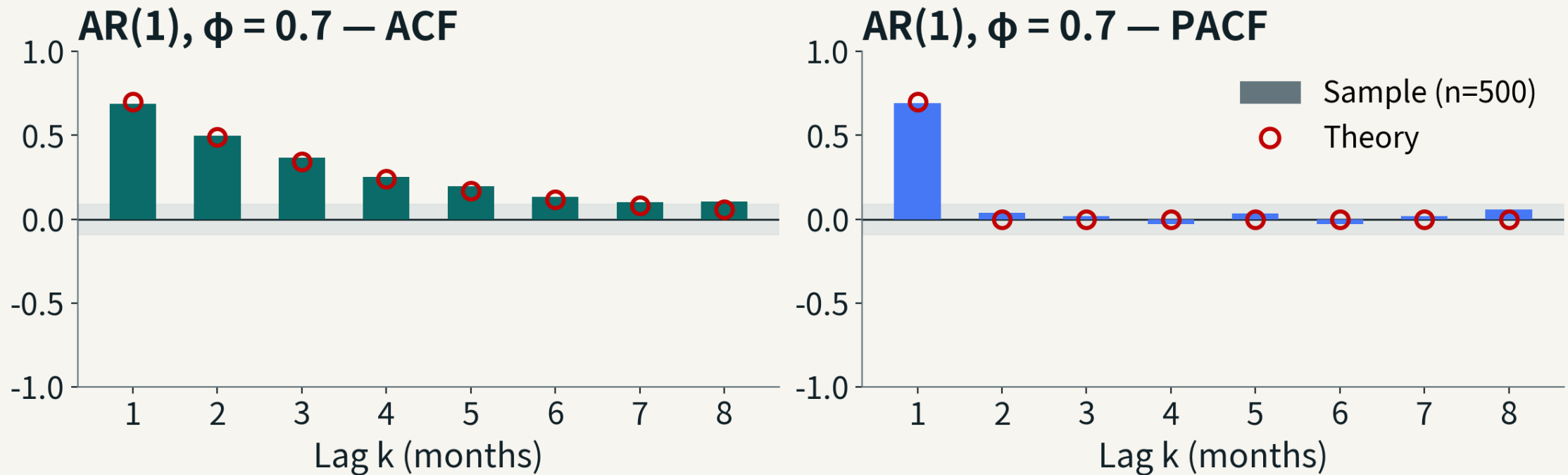
PACF cuts off after lag 1

Lag-1 value = ϕ , all others = 0

Where the PACF cuts off = the AR order

A simulation matches the theory

AR(1), $\phi = 0.7$, $n = 500$ · bars = sample, red circles = theory, grey = band



ACF decays slowly · PACF drops into the band after lag 1

04

MA(1): last month's shock lingers

Moving Average — explained by past surprises (shocks)

In MA(1), a shock lasts for exactly one extra month

$$S_t = \varepsilon_t + \theta \varepsilon_{t-1}, \quad |\theta| < 1$$

JAN

Disease at fish farms

A surprise → price jumps

FEB

Effect lingers

$\theta \times$ January's shock

MAR

Shock has faded

No effect any more

The memory of a shock is exactly one month

MA(1)'s ACF drops to zero right after lag 1

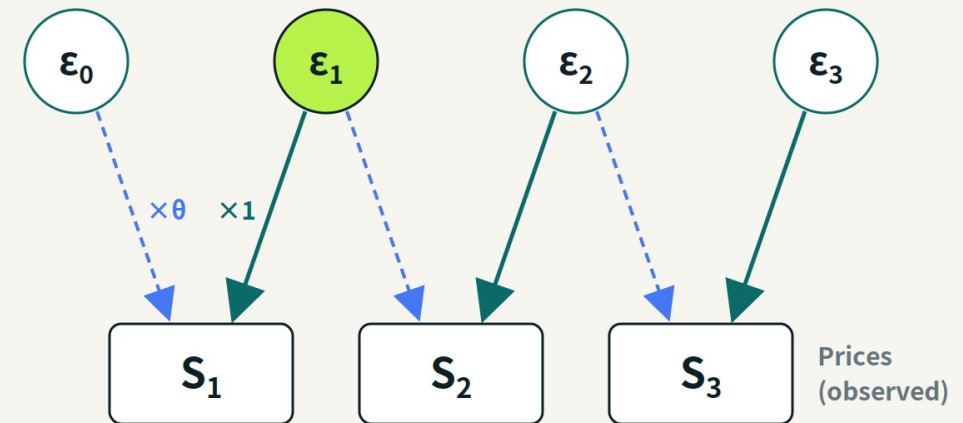
Formula

$$\rho_1 = \frac{\theta}{1 + \theta^2}, \quad \rho_k = 0 \quad (k \geq 2)$$

Example

$$\theta = 0.7 : \quad \rho_1 = \frac{0.7}{1.49} \approx 0.47$$

Shocks (not observed)



S_1 & S_2 : share $\epsilon_1 \rightarrow \rho_1 \neq 0$

S_1 & S_3 : no shared shock $\rightarrow \rho_2 = 0$

Correlation appears only when two prices share a shock

Prices two months apart share no shock, so their correlation is 0

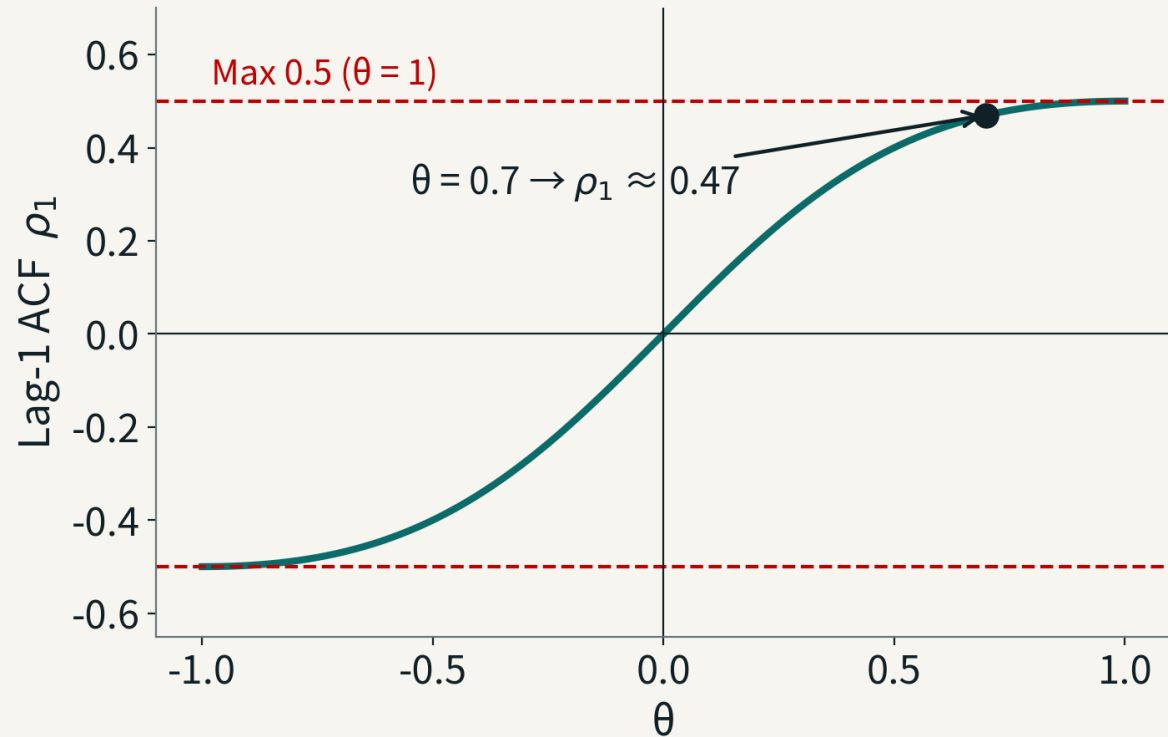
MA(1)'s lag-1 ACF can never exceed 0.5

GOOD TO KNOW

A ceiling at lag 1

- Max 0.5 at $\theta = 1$
- Negative if $\theta < 0$
- Above 0.5? Not MA(1)

A quick screen for MA(1)



Example: a lag-1 ACF of 0.8 points to AR before MA(1)

MA(1)'s PACF does not cut off; it shrinks with alternating signs

Lag-2 PACF (real lag-2 corr. = 0)

$$\phi_{22} = \frac{0 - \rho_1^2}{1 - \rho_1^2} = -\frac{\theta^2}{1 + \theta^2 + \theta^4}$$

Example

$$\theta = 0.7 : \quad \phi_{22} = -\frac{0.49}{1.73} \approx -0.28$$

PACF, $\theta = 0.7$

+0.47

-0.28

+0.19

-0.13

...

Real (0) is below the hand-off (+), so the PACF turns negative

Lag-2 PACF is the Jan–Mar correlation once Feb is known

Notation: after $|$ = already known · u, v = what Feb leaves unexplained

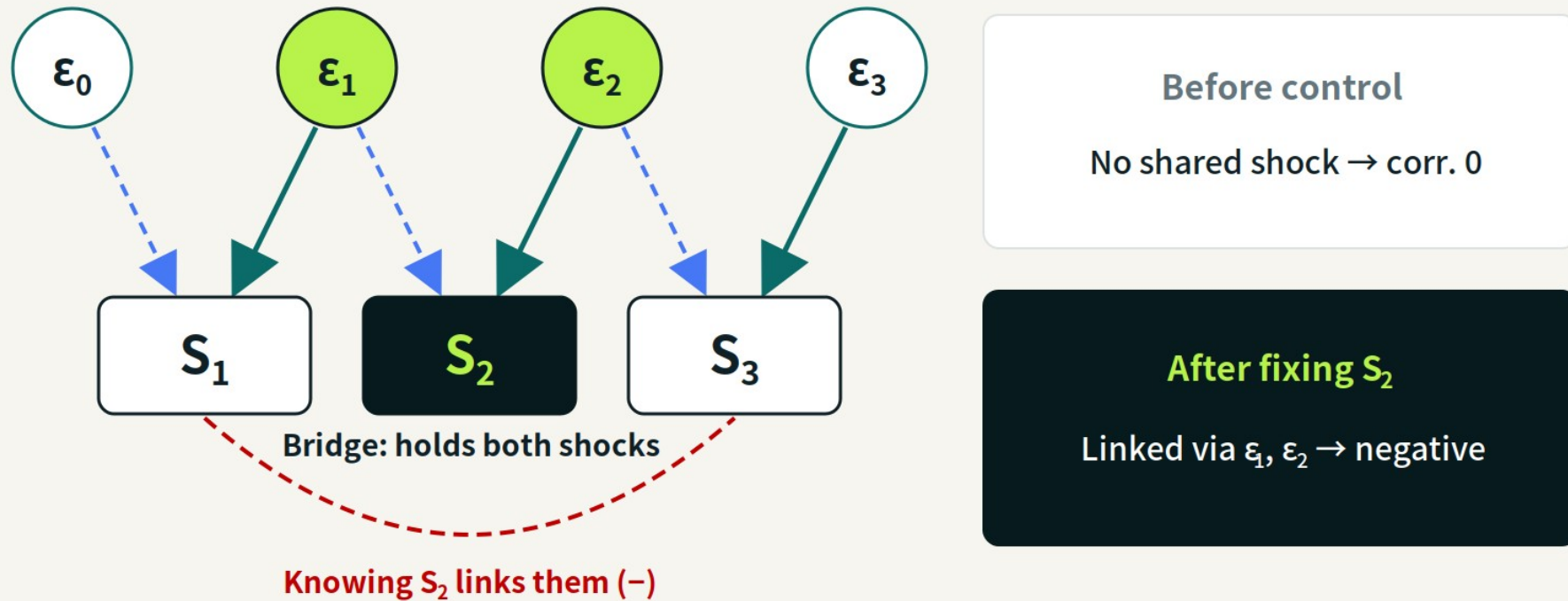
$$\phi_{22} = \text{Corr}(S_3, S_1 \mid S_2) = \text{Corr}\left(\underbrace{S_3 - \rho_1 S_2}_u, \underbrace{S_1 - \rho_1 S_2}_v\right)$$

$$S_1 = \varepsilon_1 + \theta\varepsilon_0, \quad S_2 = \varepsilon_2 + \theta\varepsilon_1, \quad S_3 = \varepsilon_3 + \theta\varepsilon_2$$

Written in shocks, Jan and Mar share nothing

So the plain correlation (ACF) is 0 — fixing Feb changes that

Feb's price is a bridge holding both Jan's and Feb's shocks



Fixing Feb mixes both shocks into the Jan and Mar residuals

A negative link appears where there was none

Writing residuals in shocks reveals what they share

u = March residual (after Feb)

$$u = \varepsilon_3 + (\theta - \rho_1) \varepsilon_2 - \rho_1 \theta \varepsilon_1$$

v = January residual (after Feb)

$$v = \theta \varepsilon_0 + (1 - \rho_1 \theta) \varepsilon_1 - \rho_1 \varepsilon_2$$

Only the shared shocks multiply through

$$\text{Cov}(u, v) = -\rho_1 \theta = -\frac{\theta^2}{1 + \theta^2}$$

$$\phi_{22} = \frac{\text{Cov}(u, v)}{\text{Var}(u)} = -\frac{\theta^2}{1 + \theta^2 + \theta^4}$$

u and v now both carry the Jan and Feb shocks (brought in by Feb)

With $\theta = 0.7$, the lag-2 PACF ≈ -0.28

Example: a high January means a lower March forecast

$\theta = 0.7$ · we already know February was above normal



Knowing a sum, a big cause implies a small other cause (explaining away)

That is why MA(1)'s lag-2 PACF is negative

If the fixed price is a cause, PACF cuts; if a mixed signal, it doesn't

AR(1)

Feb = the cause

- $\text{Mar} = \phi \times \text{Feb} + \text{new shock}$
- Knowing Feb is enough

Lag-2 PACF = 0



MA(1)

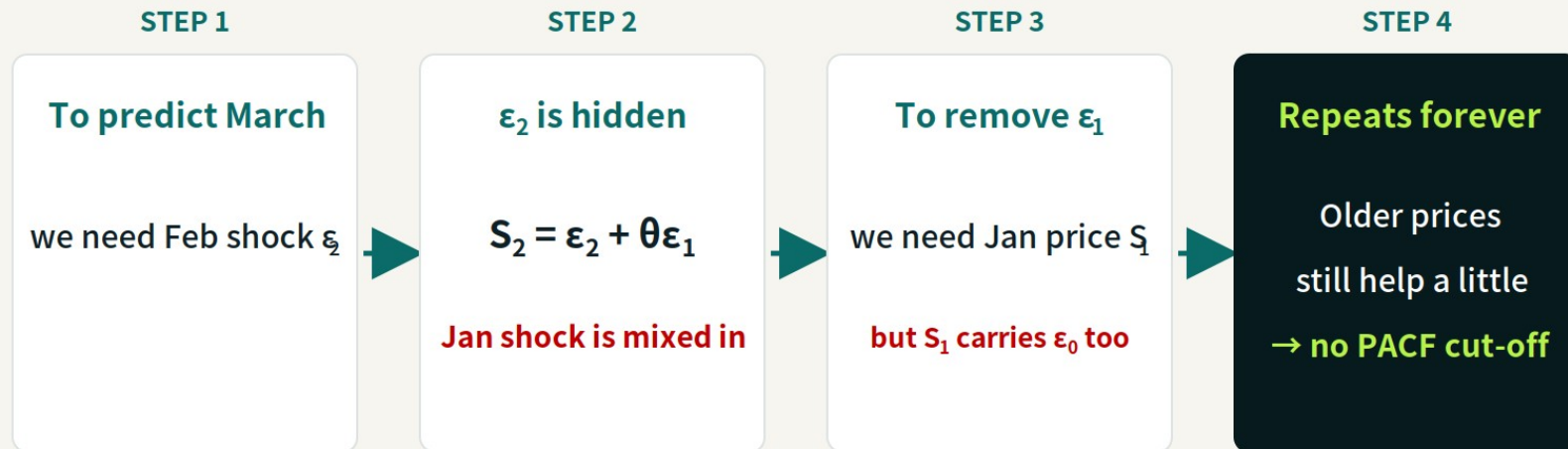
Feb = sum of shocks

- Only the total is seen
- Knowing it links them

Lag-2 PACF $\neq 0$

AR(1)'s March residual is just a new shock, unrelated to any past price

Recovering a hidden shock needs every past price



$$S_t = \theta S_{t-1} - \theta^2 S_{t-2} + \theta^3 S_{t-3} - \cdots + \varepsilon_t$$

Older prices keep direct coefficients, so the PACF decays slowly

At lag 3, January still filters leftover noise

Goal: to predict $S_4 = \varepsilon_4 + \theta\varepsilon_3$, recover the hidden ε_3

Step	Price used	What we get	Newly mixed in
1	$S_3 = \varepsilon_3 + \theta\varepsilon_2$	signal of ε_3	$+\varepsilon_2$
2	$S_2 = \varepsilon_2 + \theta\varepsilon_1$	removes ε_2	$+\varepsilon_1$
3	$S_1 = \varepsilon_1 + \theta\varepsilon_0$	removes ε_1	$+\varepsilon_0$

Using only Feb and Mar leaves Jan-shock noise; Jan's price removes more of it

So the lag-3 PACF is not zero either

More middle prices shrink the leftover noise but never remove it

$$\varepsilon_t = S_t - \theta S_{t-1} + \cdots + (-\theta)^{k-1} S_{t-k+1} + \underbrace{(-\theta)^k \varepsilon_{t-k}}_{\text{leftover}}$$

Example: leftover at lag 3

$$\varepsilon_3 \approx S_3 - \theta S_2 \quad (\text{leftover: } + \theta^2 \varepsilon_1)$$

PACF at lag k

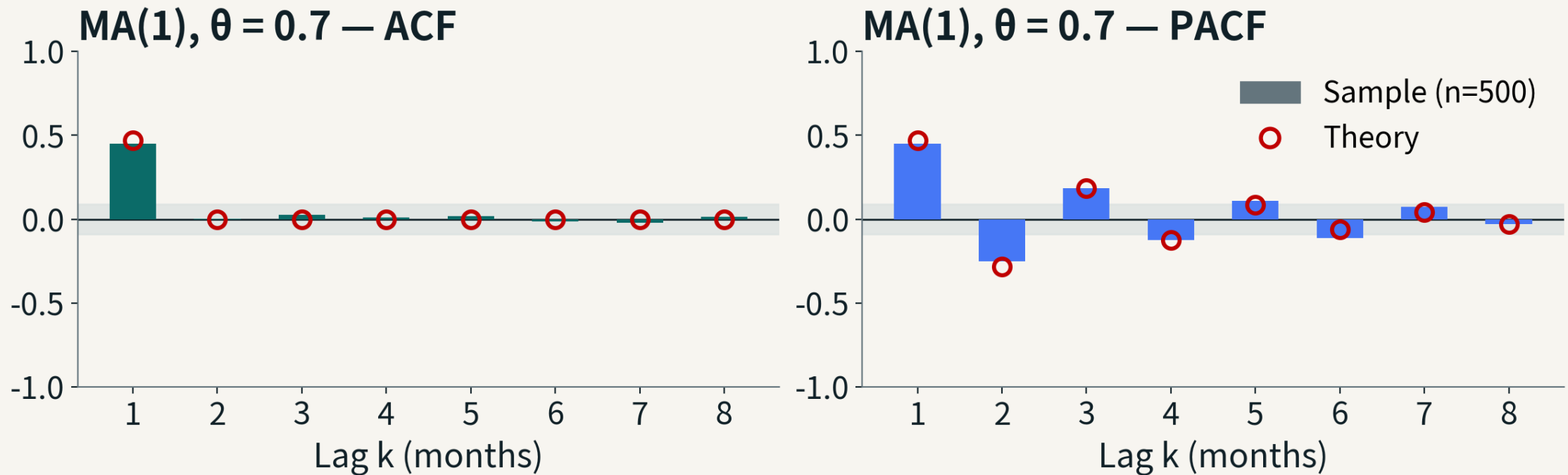
$$\phi_{kk} = -\frac{(-\theta)^k (1 - \theta^2)}{1 - \theta^{2(k+1)}}$$

Signs alternate: the recovery weights go +, −, +, ...

Sizes shrink: the leftover falls with powers of θ

MA(1) simulation also matches the theory

MA(1), $\theta = 0.7$, $n = 500$ · bars = sample, red circles = theory



ACF falls into the band after lag 1 · PACF alternates and decays

05

Picking a model

Which plot cuts off first?

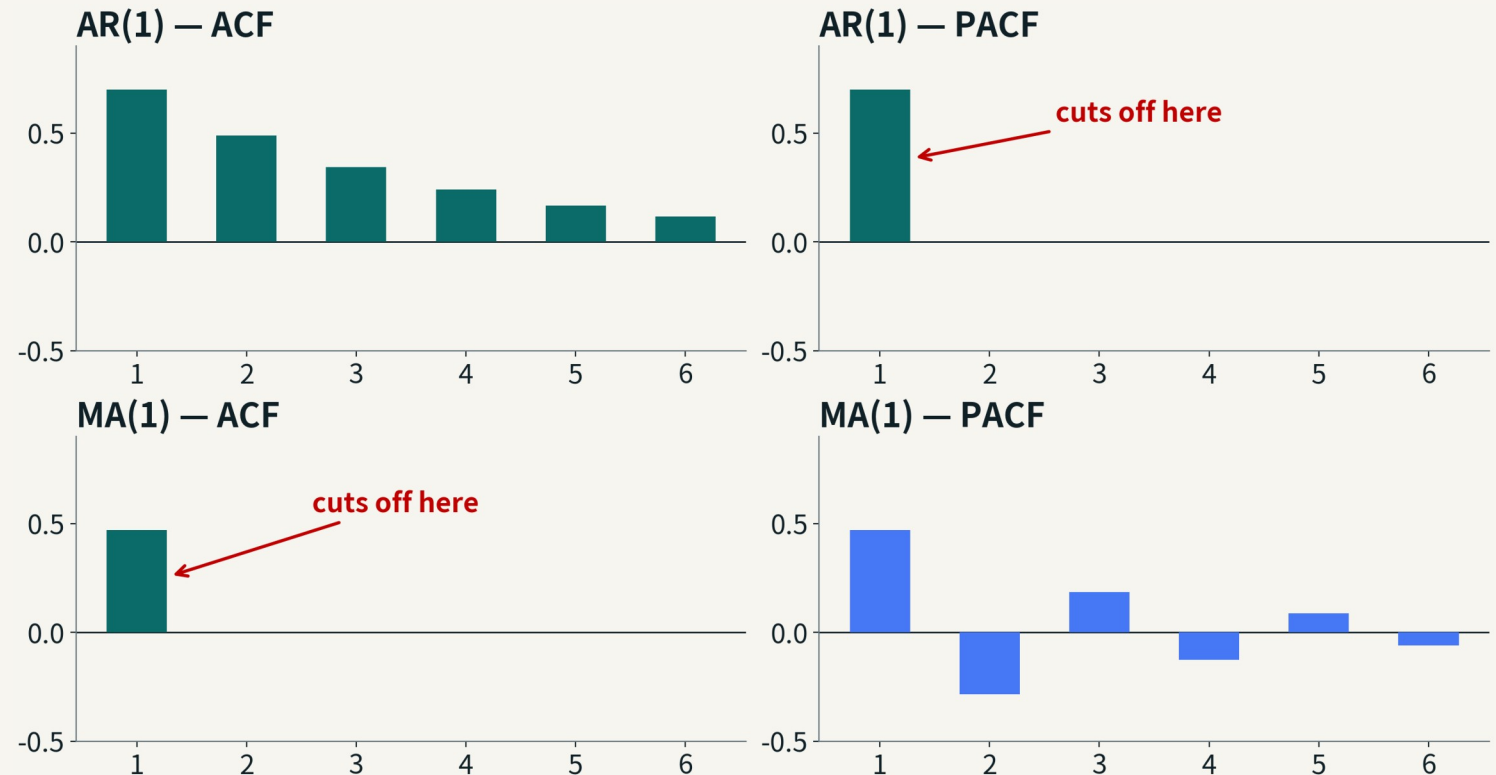
AR and MA cut off in opposite plots

AT A GLANCE

Look for the cut

- AR $\rightarrow$ PACF cuts
- MA $\rightarrow$ ACF cuts
- Opposite patterns

Cut-off lag = order



Rule: read AR order from PACF, MA order from ACF

Model	ACF	PACF	Read the order from
AR(1)	decays slowly	0 after lag 1	PACF
AR(p)	decays slowly	0 after lag p	PACF
MA(1)	0 after lag 1	decays slowly	ACF
MA(q)	0 after lag q	decays slowly	ACF
ARMA	decays slowly	decays slowly	neither cuts → compare AIC

“Prices carry on → PACF; shocks carry on → ACF”

With real data, decide in three steps



If there is a trend, difference the series first, then plot

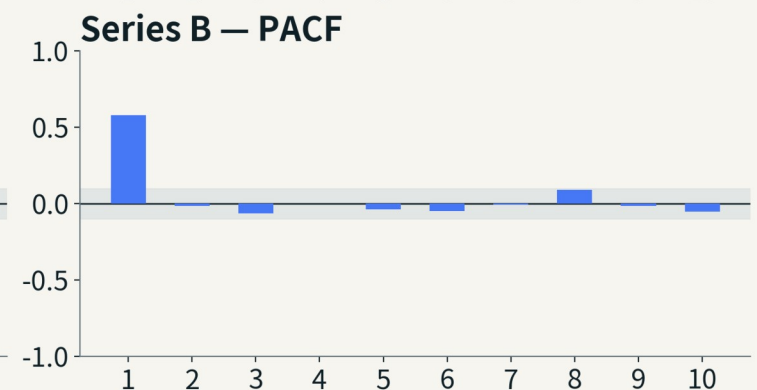
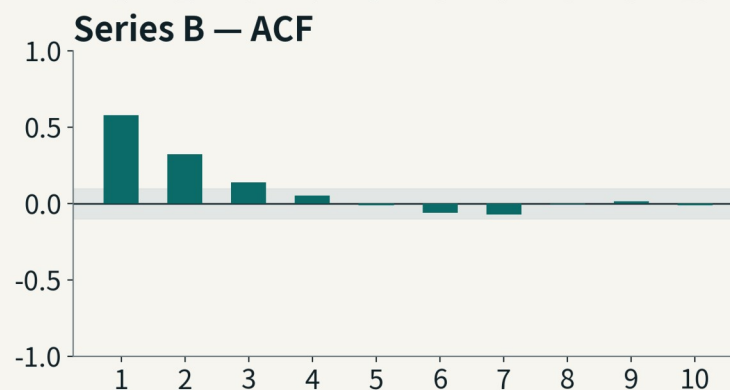
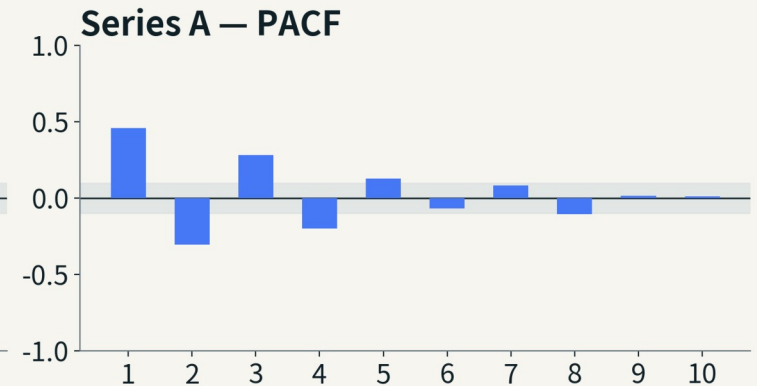
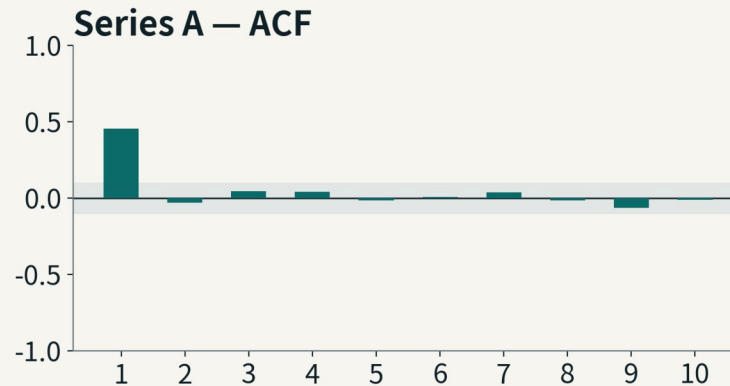
Quiz: which model is Series A, and which is B?

QUIZ

Which plot cuts first?

- Compare the two
- Find the cut lag
- Grey band = zero

Answer on next slide



Answer: A is MA(1), B is AR(1)

SERIES A

MA(1)

- ACF: 0 after lag 1
- PACF: alternating

Generated with $\theta = 0.8$

SERIES B

AR(1)

- ACF: slow decay
- PACF: 0 after lag 1

Generated with $\phi = 0.6$

Like B's lag 8, a bar can poke out of the band by chance

Judge the overall shape, not a single bar

A few lines of Python draw both plots

```
from statsmodels.tsa.arima_process import ArmaProcess
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
y = ArmaProcess([1, -0.7], [1]).generate_sample(500)    # AR(1)
plot_acf(y, lags=12); plot_pacf(y, lags=12)
```

TRY 1

Set ϕ to -0.7

ACF signs alternate

TRY 2

Switch to MA(1)

MA coefs $[1, 0.7]$

TRY 3

Try $n = 100$

Watch the band get wider

※ **ArmaProcess takes AR coefficients with flipped sign: $\phi = 0.7 \rightarrow [1, -0.7]$**

Summary: the plot that cuts off first names the model

ACF

All paths

- Direct + indirect
- MA(q): 0 after q
- AR: slow decay

PACF

Direct path

- Middle lags removed
- AR(p): 0 after p
- MA: slow decay

DECISION

Cut-off = model

- PACF cuts $\rightarrow$ AR
- ACF cuts $\rightarrow$ MA
- Both decay $\rightarrow$ ARMA

Read the pattern outside the band; the cut-off lag gives the order

ACF: total resemblance.

PACF: direct resemblance.

The first to cut off names the model.

Next time

ARMA and choosing orders with AIC

※ All price data in this deck are simulated for teaching.